

Ethernet, Fiber Optics & RJ45

Assignment Answers

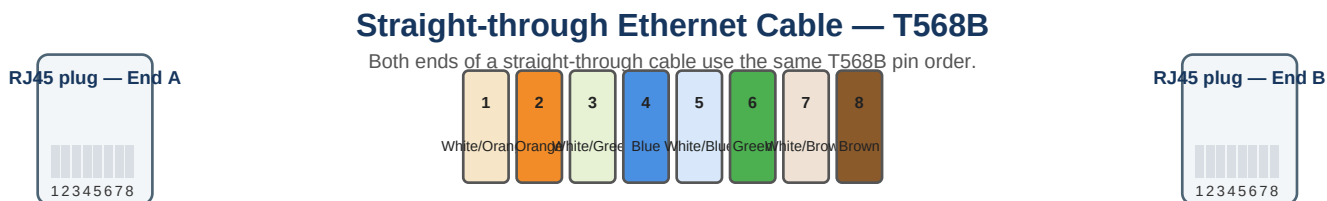
1. Ethernet cables vs. fiber-optic cables

Difference	Ethernet (copper)	Fiber optic
Transmission medium	Uses copper twisted-pair conductors and electrical signals.	Uses glass/plastic optical fiber and pulses of light.
Typical distance	Common Ethernet copper links are typically up to about 100 meters (328 feet).	Can support much longer links, from twisted pairs of meters to many kilometers.
Speed / interference	Supports high LAN speeds and is generally inexpensive.	Supports very high bandwidth but is more expensive and is immune to electromagnetic interference.
Real-world example	Connecting a home Wi-Fi router to a PC, smart TV, or network switch.	Connecting buildings in a college campus backbone or high-speed core network.

Summary: Ethernet is usually the practical choice for short, low-cost connections to computers and switches, while fiber is preferred when very high bandwidth, long distance, or immunity to electrical interference is important.

2. Straight-through Ethernet cable — T568B wiring

The T568B standard specifies the following order from pin 1 through pin 8. For a **straight-through** cable, the same order is used at both ends.



T568B pin order:

Pin 1: White/Orange Pin 2: Orange Pin 3: White/Green Pin 4: Blue
Pin 5: White/Blue Pin 6: Green Pin 7: White/Brown Pin 8: Brown

Pin sequence: 1 White/Orange, 2 Orange, 3 White/Green, 4 Blue, 5 White/Blue, 6 Green, 7 White/Brown, 8 Brown.

3. Step-by-step guide: attaching an RJ45 connector to a CAT6 cable

For this answer, I used a recent YouTube demonstration titled “**How to Crimp RJ45 Connector**” as a reference. The video demonstrates stripping CAT6, arranging the wires in T568B order, inserting them into an RJ45 plug, crimping, and testing the finished cable. ■cite■turn0youtube12■

- 1. Prepare the tools** — Get a CAT6 cable, compatible RJ45 plugs, an RJ45 crimping tool, and preferably a cable tester.
- 2. Cut the cable** — Cut the CAT6 cable to the required length and make a clean, straight cut.
- 3. Strip the outer jacket** — Use the stripper on the crimping tool or a cable stripper to remove a short section of the outer jacket. Avoid nicking the internal conductors.
- 4. Untwist and separate** — Separate the four twisted pairs and gently straighten the eight individual conductors. Keep the untwisted section as short as practical.
- 5. Arrange the wires** — Place the eight conductors in T568B order: White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.
- 6. Trim evenly** — Hold the wires tightly in the correct order and trim their ends so they are even and can reach the front of the RJ45 plug.
- 7. Insert into the RJ45 plug** — With the contacts oriented correctly for the connector, slide all eight wires fully into the plug. Check that each wire remains in its correct channel and reaches the end.
- 8. Crimp** — Place the RJ45 plug into the crimping tool and squeeze firmly so the contacts pierce the conductors and the strain-relief tab grips the cable jacket.
- 9. Inspect** — Check that the wire order is still correct, all eight conductors reach the plug end, and the jacket is held securely by the strain relief.
- 10. Test** — Use a cable tester to verify continuity and the correct pin sequence. If the cable fails, remake the connector rather than relying on an uncertain connection.

Safety/quality note: Use RJ45 connectors designed for the cable type and conductor size. Do not force a plug onto a cable if the connector is not compatible.

Video reference: [YouTube — How to Crimp RJ45 Connector](#) (Technical Th1nker). ■cite■turn0youtube12■

4. Gaming café with 10 PCs — cable and tool choice

Choice: Ethernet cable + RJ45 crimper. For a small gaming café with 10 PCs in the same room or floor, I would use CAT6 Ethernet cables because they are cost-effective, easy to install, and well suited to typical short LAN runs. I would use an RJ45 crimper to terminate custom-length copper cables with RJ45 connectors, with a cable tester used afterward to check each cable. Fiber optics would make more sense for longer backbone links or where very high bandwidth and electrical-interference immunity are important, but it would add cost and require fiber-specific tools and transceivers.

Example layout: ISP/Router → Gigabit or faster Ethernet switch → CAT6 cables → Gaming PCs 1–10.

Why: The PCs are likely to be close enough for standard copper Ethernet, so the lower equipment and installation cost is useful for a small café. Wired Ethernet also provides a stable connection for LAN multiplayer traffic and can support high-speed switching when suitable Gigabit or faster equipment is used.

Key takeaway

Use **T568B consistently** when making a straight-through cable: both ends have the same eight-wire order. Ethernet is generally practical for short LAN connections, while fiber is especially useful for longer, high-bandwidth or interference-sensitive links.